

One Coalition, Two Electorates

A short, code-free tour of what the data showed about Reform UK's voters

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MA336 · Artificial Intelligence and Machine Learning with Applications

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The question

There's a comfortable shorthand in political commentary: "the Reform voter." I wanted to know whether that single label hides more than one kind of person.

The starting idea is **cross-pressure**—a classic in electoral sociology. A voter can hold attitudes that pull against their own vote. Ford and Goodwin (2014) describe this kind of base as "left behind": socially hardline, but economically dependent on the state—a claim others have since contested (Evans and Mellon, 2016) or reframed around identity and education (Sobolewska and Ford, 2020, in *Brexitland*). Rather than assert the tension exists, I wanted to test whether it actually shows up *inside* Reform's 2024 voters—using machine learning, and using several methods so they could check each other.

Where the data made me change course

Three honest moments shaped the result, and I've left them visible rather than tidied away.

A plan the data wouldn't support. I'd intended to study LGB Reform voters and their own equality attitudes. But the survey's split-ballot design meant that of 336 Reform voters, only fifteen identified as non-heterosexual, and just two of those were asked the relevant questions. Two is not a finding. So I reframed the question from *identity-versus-vote* to *attitude-versus-attitude*—the stronger question anyway.

A judgement call over a metric. After using PCA to find the main attitudinal axes, I clustered the voters. The usual metric (the silhouette score) was highest at two clusters—but two clusters lumped almost all Reform voters together and hid exactly what I was looking for. I chose three, transparently trading a little statistical neatness for a result that answered the question. (As it happens, the two-cluster optimum is the two-electorates story; the third cluster is just a small left-libertarian fringe.)

A method that disagreed with itself. A first autoencoder seemed to confirm the clusters neatly—until I noticed its result leaned on a geometric quirk. A second one, built to remove the quirk, disagreed. Instead of quietly keeping the flattering number, I followed the disagreement, and it led to the most useful part of the analysis: pinning down exactly how much to trust the split.

The finding: two electorates, not one

Clustered on political attitudes alone, Reform's voters split almost in half:

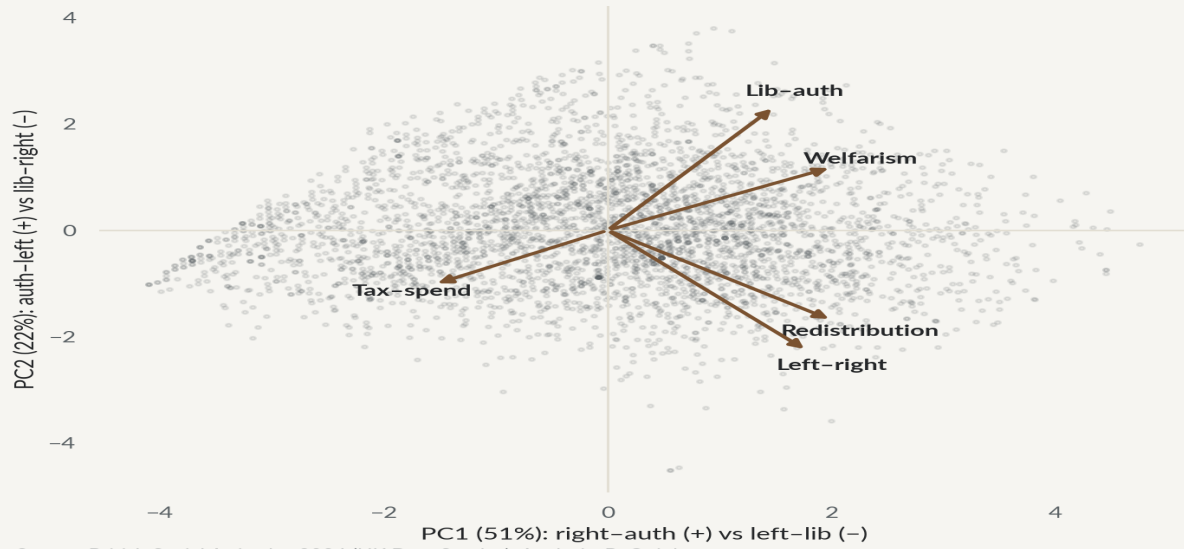
- 55% sit in a conventional **right-authoritarian** group.

- **41%** sit in an **authoritarian-left** group—socially hardline, but economically left: strongly pro-redistribution and left-leaning, and markedly the least credentialed of the three.

That second group is the cross-pressured one. Their economic instincts sit in real tension with the party they voted for—Reform campaigned in 2024 on tax cuts it costed itself at nearly £90bn a year, a figure the IFS reported and judged an underestimate (Emmerson, Joyce and Miller, 2024). But “economically left” undersells what’s actually going on. This group is pro-redistribution *and* about as sceptical of welfare claimants as the conventional right-authoritarian group *and* nearly as anti-immigration (around four in five, on the relevant split-ballot items, rate immigration bad for the economy). What makes the “left-behind” label concrete is education, not benefits: this group is much the least qualified of the three (31% hold a degree, against 45% and 64%), and that gap holds even after adjusting for social class and household income. I had expected benefit receipt to do that work, and checked it properly rather than assume—the survey’s benefit question counts the state pension, so most of the apparent gap is really age. Above 65 the two Reform groups are indistinguishable. Redistribution to an implied in-group, alongside hostility to an out-group’s claim on the same welfare state, is what political scientists call **welfare chauvinism** (Andersen and Bjørklund, 1990)—a more precise description than “a left-wing electorate inside a right-wing party.”

The two axes read as left–right and authoritarian–libertarian

PCA biplot: loadings of the five attitudinal items on the first two components

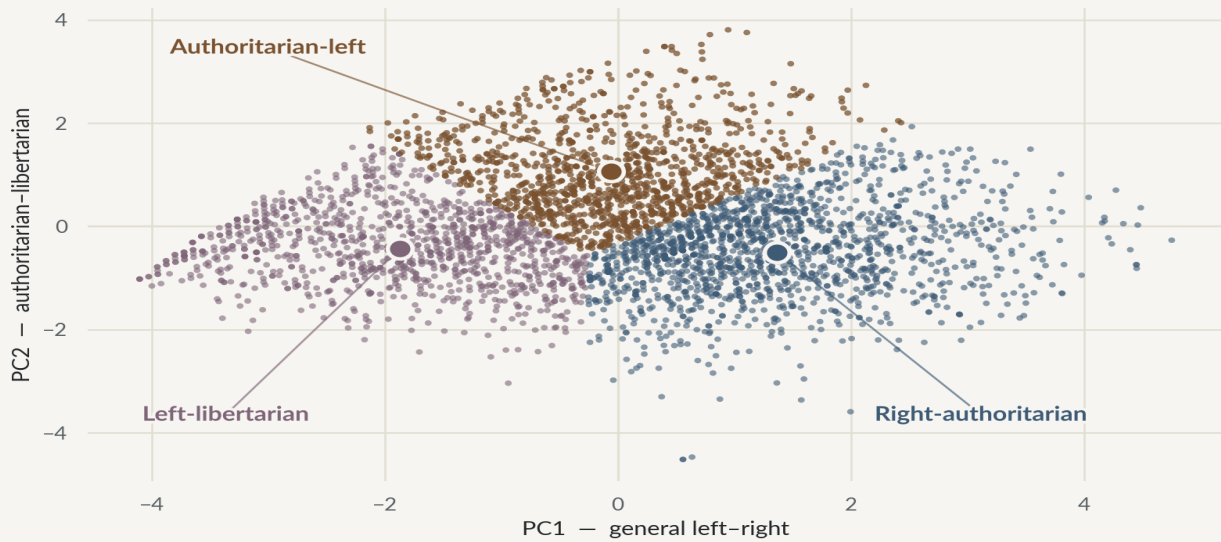


Source: British Social Attitudes 2024 (UK Data Service). Analysis: B. Quinlan.

The two attitudinal axes: PC1 (left–right fusion) and PC2 (authoritarian–left vs libertarian–right)

Three attitudinal groups in Britain's 2024 electorate

K-means clusters ($k=3$) on the first two principal components. Reform's voters concentrate in two of them.



Source: British Social Attitudes 2024 (UK Data Service). Analysis: B. Quinlan. $n=3,966$.

The three-cluster solution: ~55% right-authoritarian, ~41% authoritarian-left

Is it real—and is it old?

How much to trust it. The autoencoder that disagreed taught me where the uncertainty really lives. The split barely moves if I drop the weakest variable, cluster on the full data instead of a 2-D summary, or resample the respondents—a Reform voter placed in the left-behind group stays there across resamples 98% of the time. Hierarchical clustering agrees with the main method strongly, and a decision tree, coming at the problem backwards, finds the same dominant attitude and—tellingly—struggles to pick out the cross-pressured group, exactly as a genuinely conflicted group should. The one thing that *can* redraw the split is a radically different way of compressing the data, and even then only at the soft border between the two Reform groups, never in whether the left-behind group exists.

How old it is. Projecting 2014 UKIP identifiers into the same model, they fall into the “left-behind” cluster *even more heavily* than 2024 Reform voters—59% against 41%, a gap well beyond chance. The pattern predates Reform by a decade. Looking closer, Reform’s coalition hasn’t *moved away* from the left-behind core so much as *added* a conventional-right layer on top of it: the left-behind group is still over-represented, while the conventional-right group flips from under- to over-represented over the ten years.

What I’m claiming—and what I’m not

I’ve kept the claims honest. This is **descriptive**, the subgroups are small, and the clustering is unweighted—though reweighting the Reform split to the survey’s published weights barely changes it (52/44/4 against 55/41/4), so the two-electorates finding isn’t an artefact of that. These are not turnout predictions, and a repeated snapshot a decade apart can show the profile is durable without proving how individuals moved. But across linear and non-linear methods,

supervised and unsupervised, and across ten years of data, the same structure keeps surfacing: a single “Reform voter” label hides two quite different electorates, and the cross-pressured half is real, robust, and at least a decade old.

Works cited

Andersen, J.G. and Bjørklund, T. (1990) ‘Structural Changes and New Cleavages: the Progress Parties in Denmark and Norway’, *Acta Sociologica*, 33(3), pp. 195–217.

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Sobolewska, M. and Ford, R. (2020) *Brexitland: Identity, Diversity and the Reshaping of British Politics*. Cambridge: Cambridge University Press.

The full reference list is in [report.html](#).

Data: British Social Attitudes Survey 2024 (UK Data Service SN 9478) and 2014 (SN 7809), NatCen Social Research; English Indices of Deprivation 2019 (MHCLG); 2025 Local Elections Handbook (House of Commons Library). The raw survey microdata is licence-restricted and not included in this repository—see [data/README.md](#).